

Synoptic LE: Manual Redox Probes

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Measurements taken with SWAP Redox Probes & Calomel Ref probe

Protocol: [https://docs.google.com/document/d/](https://docs.google.com/document/d/1dL2ZPhLFBAZJmVBCfKxup_KOYY96qA5IWfJmQen50pY/edit)

[1dL2ZPhLFBAZJmVBCfKxup_KOYY96qA5IWfJmQen50pY/edit](https://docs.google.com/document/d/1dL2ZPhLFBAZJmVBCfKxup_KOYY96qA5IWfJmQen50pY/edit)

Units = mV

##Add Required Packages

##Setup

```
##### File Names - PLEASE CHANGE
#file path and name for raw summary data file
dat <- read.csv("Raw_Data/COMPASS_Synoptic_LE_Redox_2022_2025_Raw.csv")
#soildat <- read.csv("Raw Data/LE_Soil_pH.csv")

##Fix Sample Zone naming
dat <- dat %>%
  mutate(Zone = ifelse(Zone == "WE", "WC", Zone))%>%
  rename(Depth=Depth_cm)

##Std error function
std.error <- function (x, na.rm=FALSE){
  sqrt(var(x, na.rm = na.rm) / sum(!is.na(x)))}
```

Adjust Raw Readings by pH

Need to adjust for pH:

$\text{redox} = \text{redoxi} + [(\text{ph} - 5)59 + 224]$ This is the equation from Megonigal,

Patrick, & Faulkner 1993

They used a calomel electrode and used 224 to adjust to H₂,

we use a KCl ref probe, so we need to add 202 instead

```
##Remove unnecessary column in redox data
prelong <- dat %>%
  select (c(1:11), -c("Campaign", "Time", "Day"))%>%
  mutate(Notes = NA,
         Time_24hr = NA,
         Time_Zone = NA,
         Topography = NA)

#switch redox data to long format
datlong <- as.data.frame(melt(setDT(prelong), id.vars = c("Date", "Month", "Year", "Site", "Depth", "Zone"),
                           variable.name = "RP_mV"))

# ##Add in soil pH data
# all_corr <- merge(datlong, soildat, by = c("Site", "Zone"), all = TRUE)
#
# ##Adjust redox for pH
# all_corr <- all_corr %>%
#   mutate(value_corr = value + ((pH - 5)*59 + 202))

##Adjust redox for KCl ref probe
datlong <- datlong %>%
  mutate(value = as.numeric(value),
         value_corr = value + 202)

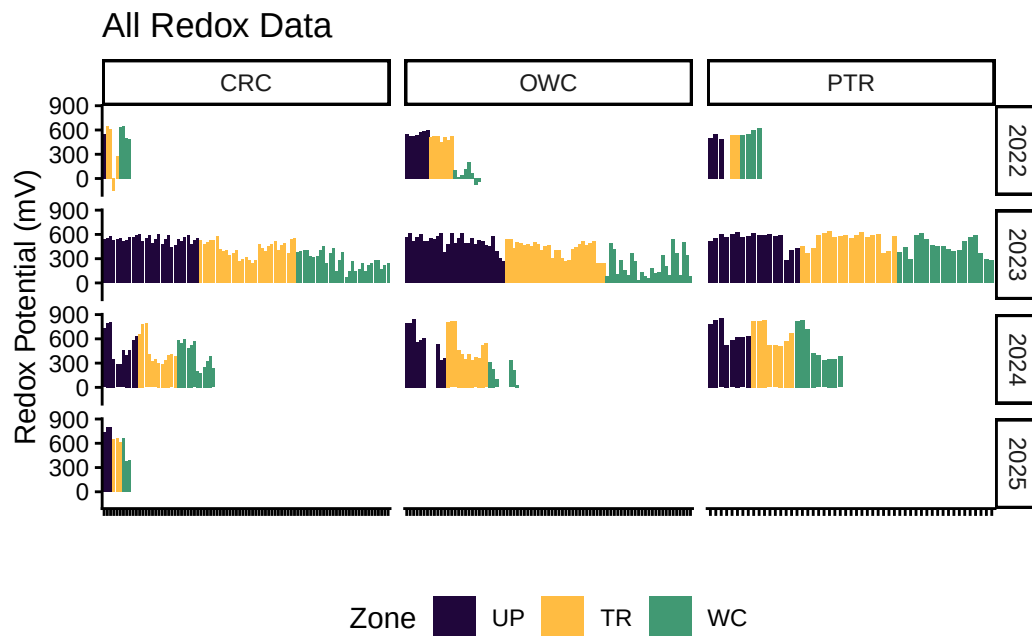
##Average across all 5-6 measurements for each sampling Site, Zone, Depth, Month, Year
red1 <- datlong %>%
```

```
group_by(Date, Month, Year, Site, Zone, Depth, Time_24hr, Time_Zone, Topography, Notes) %>%
  summarise(Date = first(Date),
            Avg_mV = mean(value_corr, na.rm=TRUE),
            Std_dv = sd(value_corr, na.rm=TRUE),
            Std_err = std.error(value_corr, na.rm=TRUE),
            Notes = first(Notes)
  )

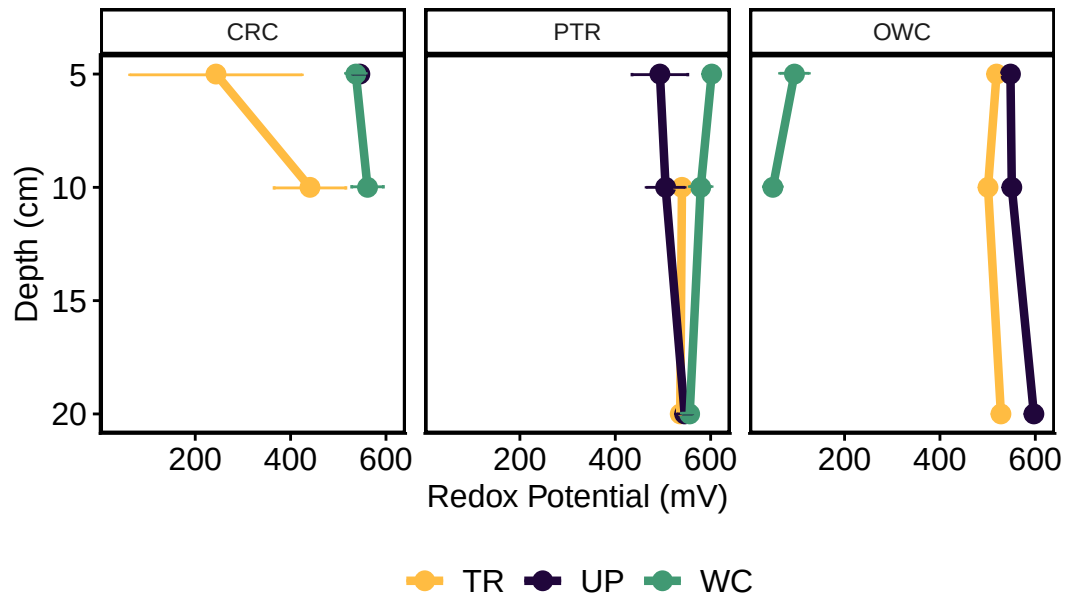
# head(red1)
```

##Write Out Adjusted & Avg'd Data

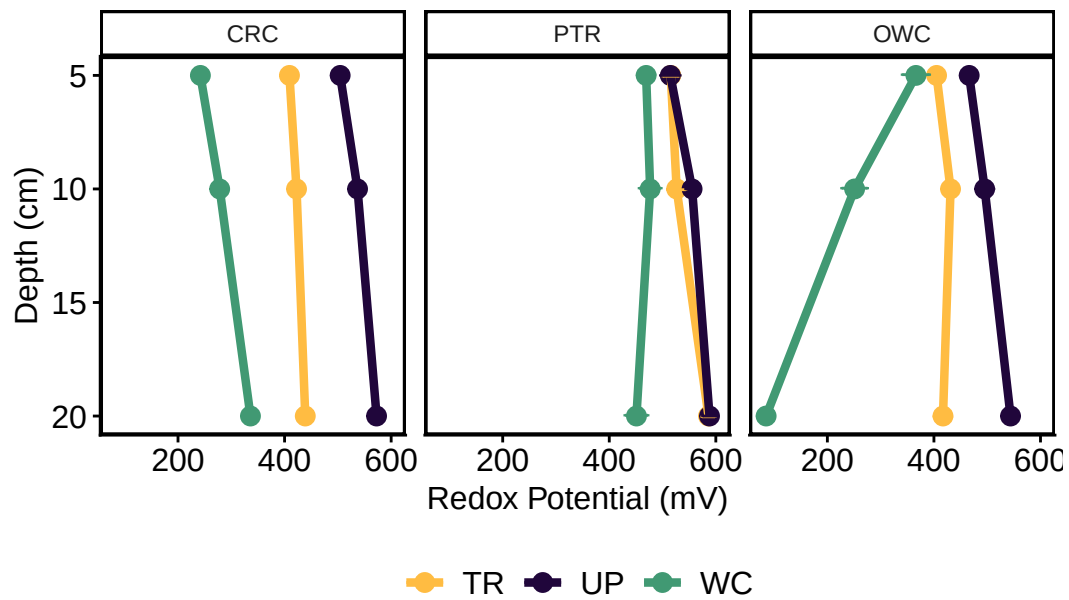
Visualize Data



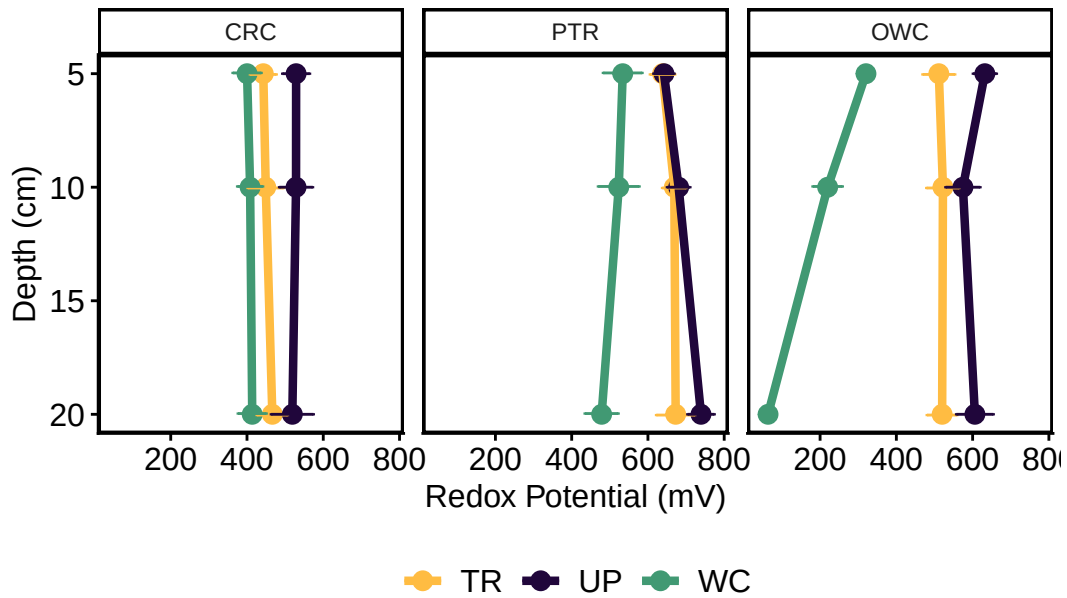
Averaged Redox Data — 2022



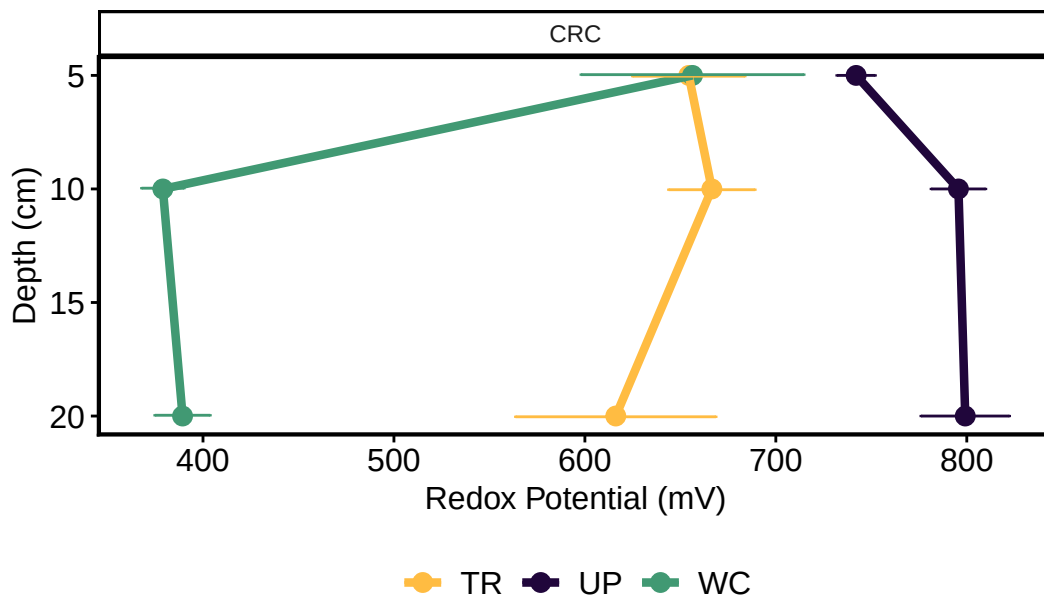
Averaged Redox Data — 2023



Averaged Redox Data — 2024

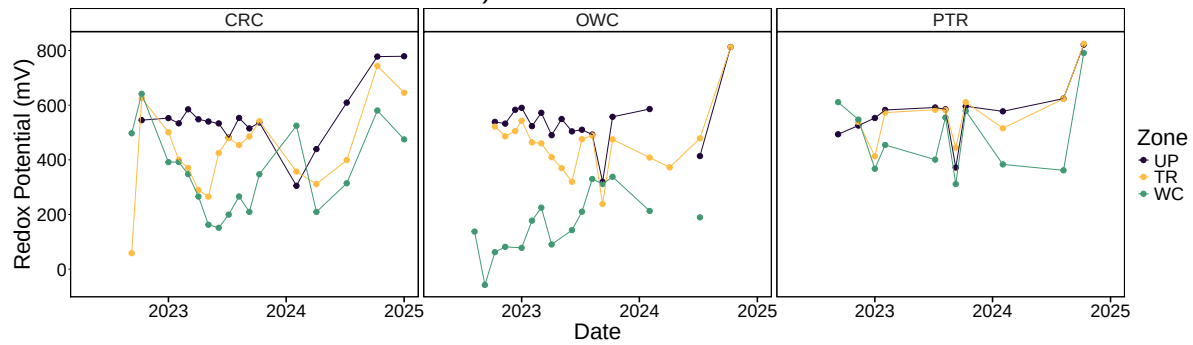


Averaged Redox Data — 2025



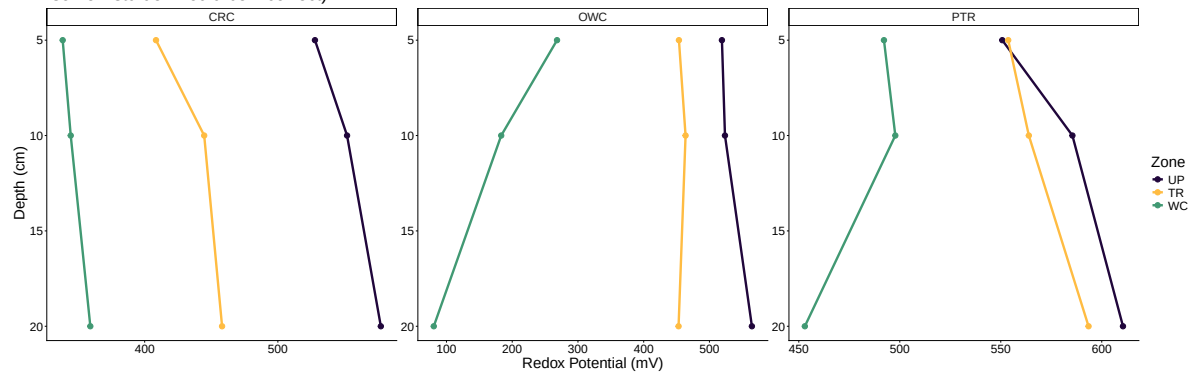
Summarized data for Site and Zone

Samples: Redox
(No std dev because values were averaged previously,
so new std dev would be incorrect)

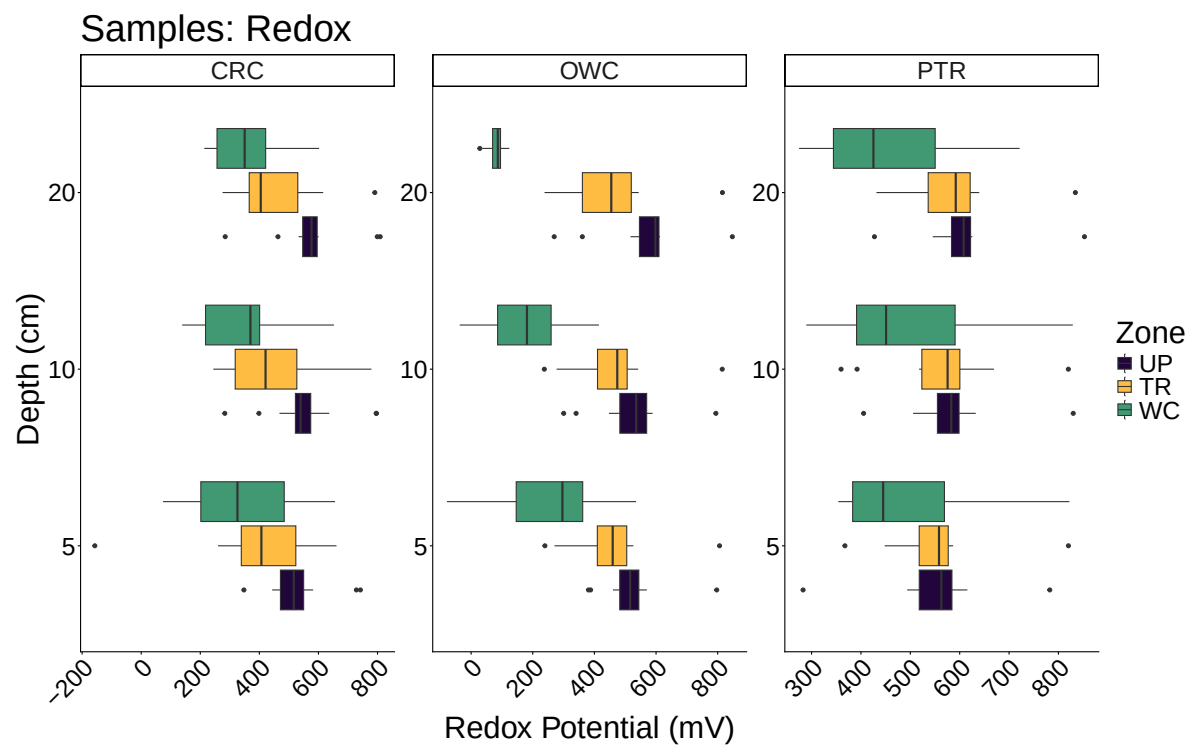


Summarized Data by Depth

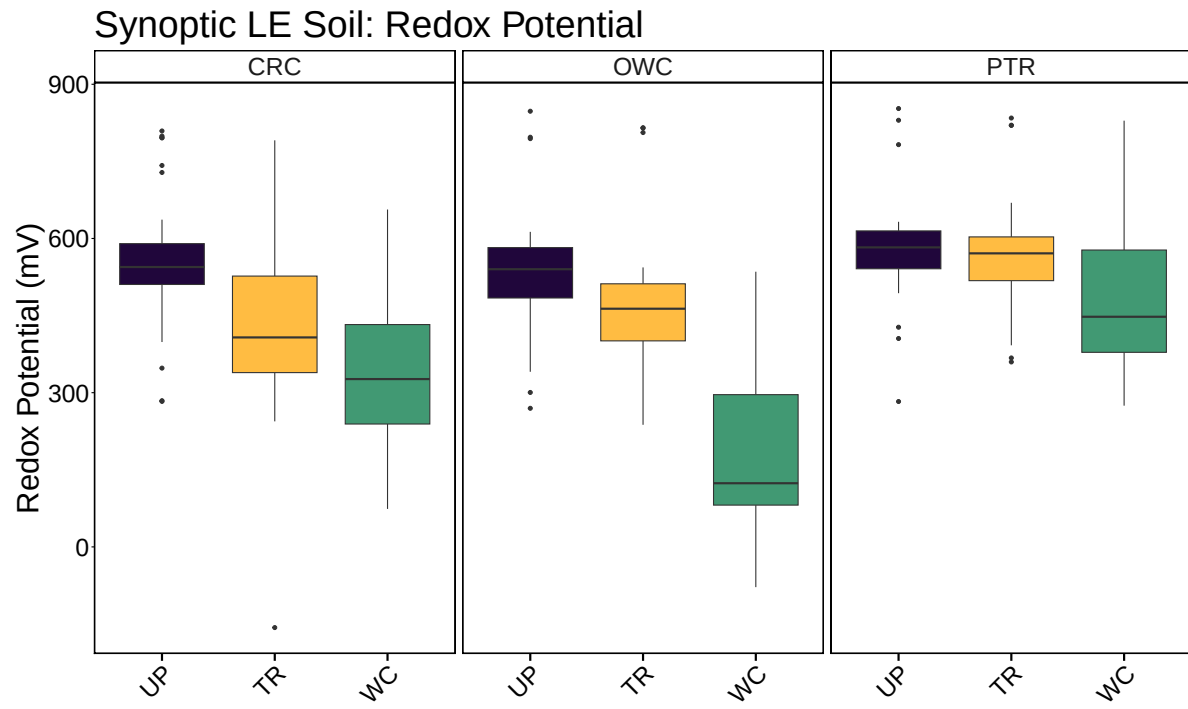
Samples: Redox
(No std dev because values were averaged previously,
so new std dev would be incorrect)



Boxplot by depth



Boxplot summary



pdf
2

##Calculate summary metrics
##Calculate number of samples

Table 1: Summary Statistics

Site	Plot	Min (mV)	Max (mV)	Mean (mV)
CRC	TR	-156.97	790.95	436.25
CRC	UP	283.05	809.15	551.27
CRC	WC	73.90	656.37	346.64
OWC	TR	237.53	815.43	457.00
OWC	UP	269.65	847.57	534.83
OWC	WC	-78.30	535.33	184.81
PTR	TR	359.80	834.32	570.12
PTR	UP	282.80	852.77	580.22
PTR	WC	274.73	829.18	481.52

Table 2: Total Samples: 368

Year	Number_of_samples
2022	41
2023	225
2024	93
2025	9

Summary Tables

Number of Samples

